Research statement

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Math Brain and its open-source framework: [1, 23, 54]

I am a mathematician specializing in probability theory, random matrices, and the mathematics of deep learning. I completed my PhD at Penn State under the supervision of Leonid Berlyand, and am now a postdoctoral fellow at the Hebrew University of Jerusalem, hosted by Sasha Sodin and Amit Daniely. My research connects a common set of questions across disciplines: how information moves through networks, how memories are organized, and how reasoning produces new knowledge. I investigate these questions through mathematical research, AI memory and agentic systems, and the tracing of flows through networks (e.g., memory flows and cryptocurrency flows). These projects inform one another. Methods for tracking flows can suggest more efficient ways to navigate a memory network, while AI systems can help identify mathematical connections and formulate new conjectures. My interest in biology currently centers on understanding memory and reasoning in human thought, using those ideas to develop AI systems that may, in turn, offer mathematical models for studying cognition. My long-term goal is to develop the mathematical foundations of general-purpose reasoning and discovery systems, with an emphasis on the problem-solving mechanisms that can advance research across many disciplines. My longer-term interests include understanding cognition, the mechanisms underlying AI and its capacity for autonomous action, the biological processes that drive human aging, and the computational principles that connect these questions.

My main current research project is Math Brain, a persistent memory and coordination system for AI-assisted mathematical research [1, 23], supported by an open-source Brain framework [54]. It organizes claims, evidence, methods, and connections into a shared knowledge network, with tools for retrieval, memory consolidation, and coordination among agents. I use it both to investigate mathematical problems and to study how accumulated knowledge affects reasoning, discovery, and autonomous problem solving.

The broader research program brings together coordinated AI agents working on complementary tasks. Some develop and test memory and retrieval systems; others investigate cryptocurrency flows, network algorithms, and open questions in probability theory and deep learning. These projects let methods move between fields while testing whether agents can build on one another's work. Participation in Kaggle competitions, including geometric packing and quantum-circuit compilation, provides concrete settings for evaluating planning, experimentation, and learning from feedback [55]. Historical and educational video production explores how AI systems can explain complex ideas and make their work accessible to people [56]. Across these activities, security audits, independent checks, and comparisons against simpler methods help show which approaches work and which need revision.

The work so far spans more than a million Claude model calls and more than 17,000 Codex sessions. The Brain has grown into a research network of more than 366,000 nodes and 16 million connections, linking mathematical results and proof strategies with memories, code, tools, experimental findings, and lessons from previous work. The overall goal is agency: enabling agents to carry a hard problem further before a person needs to step in. Recall, finding the right node when an agent needs it, supports that goal, but it is only the first layer. The Brain is also a mathematical object that I study using random walks, spectral theory, and random matrices, investigating how its structure affects the ability to retrieve knowledge, connect ideas, and make progress on difficult problems [23].

This program grows out of a longstanding interest in AI, probability theory, network science, and the connections between them, including my doctoral work on the mathematics of deep learning. Part of my role is to understand how advances in AI change the problems we can investigate. The steep historical fall in inference costs [57] makes it important to reassess capabilities rather than assume today's limits will remain fixed. Over the next two to three years, I want to keep the Brain current with the state of the art, testing agents against curated suites of open mathematical problems, computational and optimization problems, cryptocurrency-flow tasks, and competition problems. I will use these evaluations both to study problem

solving and to identify weaknesses in the Brain's own retrieval, memory, and coordination mechanisms, then test whether proposed improvements help agents work more efficiently and independently.

Over the following decade, I am interested in computational efficiency and resource allocation, as well as the biological mechanisms of memory, reasoning, and aging. More efficient computation could reduce the energy, water, and other resources needed to obtain useful results, improve the use of data centers, and make substantial computational capabilities accessible to more people. With collaborators, I want to develop models that clarify biological and cognitive processes and their connections to computation, while also studying networks of resource distribution. I also want to develop ways for systems to explain what they learn so that people can guide their development and make informed personal and collective decisions.

Looking twenty years ahead is much harder, but my ambition is to develop reliable systems that help people solve difficult problems across science and practical life, while keeping their reasoning understandable. As capabilities grow, human judgment, democratic institutions, and public participation must remain central to what these systems pursue and how they are used. I want to develop ways for people to inspect these systems' evidence, question their recommendations, and revise or stop their actions. Alongside the technical work, I aim to make their capabilities and limits understandable to the public, so that decisions about their use are shaped by informed democratic participation and institutions that remain accountable to people.

1. Why agents need a memory

Often when agents do research they get stuck in a loop where they keep doing the same thing over and over again. An agent proves a lemma it already proved last week or tries a route the agent next to it just gave up on, and when its working context fills up and is wiped it can lose a theorem. A research group does not work that way, because people remember what was tried and why it failed. I want agents to have that memory and to reach for it at the right moment. So I count how far an agent gets on a fixed budget and how often it repeats a step already on record (and what a right answer costs).

Memory is also essential for learning from mistakes. A substantial part of the system is devoted to recording errors identified by agents or through my corrections, together with the circumstances that produced them and the checks that could have prevented them. These records can become contextual triggers embedded in the workflow: when an agent encounters a related problem, tool, or action, the relevant warning is surfaced before it proceeds. The aim is a practical form of learned caution, in which experience changes how the agent approaches a familiar risk. These warnings remain evidence to assess, and their relevance must itself be tested, so that remembering a mistake helps prevent its recurrence without turning an earlier lesson into a new source of error.

2. What the brain is

There are two stores, one inside the other. The inner store, Math Brain, organizes mathematical knowledge into typed cards: definitions, theorems, lemmas, proof techniques, worked proofs, counterexamples, computations, open problems, and unsuccessful proof strategies. Each card carries its sources and verification status, distinguishing established results from provisional, refuted, or superseded claims. Links record relationships such as 'depends on,' 'proves,' 'refutes,' 'generalizes,' and 'cites.' When a claim is corrected, the earlier record is retained and linked to its replacement, preserving both the error and the lesson. The current public copy contains 86,589 cards connected by approximately 119,500 active typed links [1].

The research brain is the larger private store around it. It holds more than 30,000 written memories (results, rules, mistakes with their mechanism), together with the cards, prompts, tools, files, keywords, reflexes and hooks. A reflex is a recorded mistake compiled into a trigger and a rule. Agents write some of its edges when they connect two results, and the rest build up by themselves. When agents use two nodes in one action, or one right after the other, the edge between them gets stronger, and when they stop it fades. So agents use a theorem, a proposition or a tool a few times and all of a sudden there is a connection. That itself teaches agents what is connected, what pieces of code, what keywords, when a theorem and a piece of code go together.

The brain is not just an index. It is an index together with mechanisms that improve an agent's ability to push proofs forward and do research without getting stuck in loops. The network sits in the background like a subconscious, and the hard part is getting what it holds into the agent's context when it is needed. Recall is what an agent asks for, but an agent cannot search for what it does not know it needs. So a layer of hooks (small programs that run on every prompt and tool call) puts the nearest cards, and the mistakes recorded in that territory, in front of the agent unasked. Agents also write new hooks, skills and workflows (written procedures another agent can load and follow) as the brain develops. So the entire thing is a growth mechanism where agents can work faster, recall better and push things forward while the work itself adds to what they recall from. The store was built the same way. Thousands of agent sessions, directed by me, read the literature of probability and deep learning, wrote the cards, proved and refuted what they could, and connected results across fields.

A central part of that workflow is the frontier mechanism, which records where each problem currently stands: the strongest results, tried approaches, and outstanding obstacles. Agents consult these up-to-date records before starting work to identify the cutting edge of the problem. They then add new findings or failures as work proceeds, making the next starting point available to other agents. This keeps research moving forward instead of rediscovering results or returning to routes whose limits are already known.

Testing is one of the most important parts of this project, and designing meaningful evaluations occupies much of my time. I test recall and sustained agency through tasks ranging from locating nested files in repositories or storage archives to multi-stage scavenger hunts. As an illustrative design, a poem can encode a rule for constructing a graph. The agent must infer the rule, write code to generate the graph, and compute an invariant within a fixed time and memory budget. That value identifies a document in a repository, which points to a storage archive where the relevant file has to be distinguished from plausible but outdated alternatives. Later stages can require combining discoveries from several earlier steps, checking a mathematical argument, or handing the task to another agent through a saved record. Extending such a design to fifty steps would probe whether an agent can sustain a coherent line of work across tools, interruptions, and shifting contexts.

Mistakes are part of what the evaluation is designed to reveal. An agent might misread the poem, build the wrong graph, or obtain a plausible number from code with an indexing error. Intermediate checks expose inconsistencies without supplying the answer, prompting the agent to test assumptions, run small cases, and revise the approach. It records the failed interpretation, the cause, and the correction so later work can build on it. When a similar situation appears again, the system can surface the earlier lesson as a warning and the agent must assess its relevance. By comparing otherwise matched runs with and without these mechanisms, I can measure completion, repeated errors, recovery time, computational cost, and the level of human assistance required.

Auditing is another central part of this process. The Brain gives reviewing agents access to a result's sources, assumptions, proof attempts, code, and earlier corrections, so they can examine how a conclusion was reached. A reviewer can check a proof step by step, rerun a computation, look for counterexamples, or test whether a claim reaches beyond its supporting evidence. Because agents can share the same mistaken assumptions, I emphasize checks that use different methods and distinguish agreement between agents from independent verification. Audit findings are recorded alongside the original work, with unresolved objections kept visible and corrections linked to the claims they replace. The aim is to make errors discoverable and their lessons reusable as the research grows.

3. The mathematics of the brain

One mathematical approach I investigate is graph-based recall, where information linked to a query spreads through weighted connections to surface related knowledge. As a simple example, represent the store as $G = (V, E, w)$, with cards V, links E and weights $w(i, j) > 0$. Let A be the adjacency matrix, D the diagonal matrix of row sums and $P = D^{-1}A$, so that row i of P says what fraction of card i's weight goes to each neighbour. A question q gives each card two scores, the BM25 score $\ell_i(q)$ of its words

(the standard word-match score [2], in which rare words count for more) and the cosine $m_i(q)$ between embedding vectors, which is high when two texts say the same thing in different words. After standardizing each over the candidate cards, the seed on the research brain is $s = 0.7\,z(\ell) + 0.3\,z(m)$, with $s = 0$ off the candidates (the math brain adds scores for symbols and formulas), and activation can then spread along the links for at most two hops, with fixed weights β_1, β_2:

$$a(q) = s(q) + \beta_1 P^{\mathsf{T}} s(q) + \beta_2\,(P^{\mathsf{T}})^2 s(q).$$

The entry $(P^{\mathsf{T}}s)_i = \sum_j P_{ji}s_j$ is what card i collects from the cards that link to it, so a card the question's words never touch can still enter the pool if well-matched cards point at it. When $\beta_k = c^k$ these are, up to the factor $1 - c$, the first terms of the personalized PageRank series $\sum_k c^k (P^{\mathsf{T}})^k s$ [3, 4]. That is the model. The code uses A^{T} in place of P^{T}, with 0.55 per link, because dividing by the row sums changed first place on only one question of 381. Undivided, a card passes its whole score to every neighbour, and many steps would concentrate the mass on the leading eigenvector (eigenvector centrality), whatever the question was. Two steps hurt us in a different way. Sixteen junk cards, 1.4 percent of an early store, were not hubs (degree 3 to 5), but 92 percent of their edges stayed inside the group. They kept feeding each other and held the top three places on 39 percent of the questions the store could answer. We took the junk out of the seeds and turned the walk off.

On my stores so far the walk mostly reorders what the words already found. With a synthetic new memory on the research brain's live graph and real prompts, spreading from the ten best word matches never put it in the top ten, in 7,440 trials at each of 1 to 20 edges. The links got it into the pool and no further. On real cards the walk probably adds little because linked cards have about ten times the word overlap of random pairs when rare words count for more. Recall now runs without spreading by default and the agent walks the links by hand when it wants to (very cheap, see section 4). I doubt this is general. DAPS [5] reports strong recall on held-out Mathlib theorems with an encoder over typed dependencies, and I want a theorem that says when links help.

The learned edges of section 2 form a second graph, the use graph. A node's weight is its base-level activation in the sense of ACT-R, a standard model of human memory [6, 7], and an edge's weight is a count of uses together, read with a decay:

$$B_i(t) = \ln \sum_k (t - t_{ik})^{-d}, \qquad w_{ij}(t) = \sum_k e^{-(t - t_{ijk})/\tau},$$

where t_{ik} are the past times node i was used and t_{ijk} the past times i and j were used together, and d and τ set how fast each fades. With $d = \frac{1}{2}$ and time in days, one use yesterday contributes 1 and ten uses a year ago contribute $10/\sqrt{365} \approx 0.52$, so the memory touched yesterday outranks the one that was popular last year.

The hooks of section 2 also have to know when to stay quiet. If a hook writes a note on every prompt the agents learn to ignore it. Let $a_{(1)} \geq a_{(2)}$ be the two largest scores for a prompt, and serve a note only if $a_{(1)} \geq \theta$ and $a_{(1)} - a_{(2)} \geq \gamma$. A threshold on the word score alone already separates 2,000 questions the store can answer from 60 it cannot, with AUC 0.997 and 6.4 percent of the answerable ones wrongly getting no note [1]. New cards have to pass a test too. Let $\mathrm{acc}(S)$ be the fraction of a held-out bank of questions whose answer card ranks in the first eight when recall runs over the store S. A packet Π of new cards enters only if $\mathrm{acc}(S \cup \Pi) \geq \mathrm{acc}(S)$ on a copy, so growth cannot lower the score on that bank.

To stop the loops of section 1 we built the frontier layer. Each open problem agents are working on has a file with the statement, the latest papers and the step where their proofs fail, what is settled, which routes are dead and why, and the next targets in order. An agent starts from that failing step and checks its route against the settled facts and the dead routes. If it repeats a dead route, the agent is stopped unless it writes a concrete reason its route differs. On 28 October the layer covered 64 problems and 2,623 routes. In a preregistered test with a blind judge on two open problems, eight comparisons in all, a plan written with a digest of what the store held on that problem (the digest's second version) was preferred over a plan

written without it 7 of 8 times and had 8 redundant or dead-end steps against 17 [1]. That test judged plans and not proofs, and whether the layer gets further on hard problems is the main test of next year.

Speed matters too, because agents stop calling a memory that is slow. When agents measured recall in October, the gains came from not starting over on each call and none came from ranking. A resident server took a recall from 26.8 to 7.4 seconds at the median, with identical answers [1].

4. The tests

Every test compares setups (arms) on the same questions with the same model and time cap, and the p values are exact binomial tests on the pairs where exactly one arm was right. The table's first test has 988 items, lookups over the store's records, with answer keys made by script, and proof audits, where an agent names a proof's first bad step or passes it. There the memory arm had recall and the hooks and no file tools, the repositories arm could read, list and search the research repositories that hold the same records, and the bare arm had the model alone. In the second, the git test, both arms had the repositories and one of them also had recall. The public repository has the design and aggregates of every test, including those that showed no gain [1].

Test	Agents	Other arm	Memory	Discordant, p
988 items from a bank of 923 scripted lookups and 77 proof audits, 600 s each	Claude	82.0%	95.4%	146 to 13, 1.2e−29
2,000 questions on files in 113 git repositories, one recall a question (both arms reached 1,726)	Codex	91.4%	93.7%	83 to 43, 4.6e−4
160 proof audits with planted faults, the brain's notes against the same kind of notes scrambled, replicated on a second bank		117	128	18 to 7, 0.043

On the 988 items the bare arm got 18.9 percent. The memory arm took about half the time of the repositories arm (120 seconds an item against 211) and timed out a fifth as often (3.7 percent against 17.5). Its finished sessions read 20,000 tokens against 80,000, and a right answer cost $0.074 against $0.153 (counting a second pass with a longer limit). On the 806 items both arms finished they were tied, 99.3 and 99.4 percent (a set selected by finishing). So the gain is more questions finished on the same budget, which was the whole idea in section 1. The git test is harder for a memory, because both arms have the files and the questions are about those files. Where both arms committed to an answer they were right about equally often. The lead comes from the repositories arm giving up more often. The memory arm was slower, by about one recall on a busy machine (a median of 67 seconds against 13). The git scores come from a regrade done after the answers were known, under three rules applied to both arms.

The links and the notes did not raise the score, but they cut the cost. On 100 questions on results in the store, agents that also got the hook's notes and could step from card to card along the links scored 82 against 83 for recall alone, on 7.6 times fewer input tokens (1.8 times fewer uncached) and 78 session-minutes against 666 [1].

Recent equal-information evaluations test both retrieval and sustained work. On a 300-question recall bank, a Codex agent using served Brain recall identified the correct memory on 214 questions, against 168 for ripgrep over the same memory folder with the same file reader (68 versus 22 discordant pairs, $p = 1.2\times10^{-6}$) [60]. This was a development and tuning bank, not a fresh demonstration of generalization: two subsequent 120-question checks scored 49 versus 47 and 42 versus 48. It is a recall result, not a scavenger-hunt completion result.

Multi-day continuity pilots give a different kind of evidence [59]. Agents resumed work from a journal and the same complete records of earlier sessions, locating relevant information among stale same-topic material and combining records across days. In pilot v19, the partial-Brain retrieval arm answered both target questions correctly within a predeclared 12,000-reading-character budget on 8 of 12 test runs, against 1 of 12 for grep. In the padded stale-record pilot v21, at an 8,000-character budget, the corresponding counts were 6 of 12 and 0 of 12. Each pilot comprised two repetitions of six reused development tasks. All arms eventually answered every task correctly without the reading-budget condition. These pilots used

scratch retrieval systems with lexical and vector search, not the full use-grown graph or its hooks. The advantage was selective reading, not lower overall computational cost: the retrieval arm used about 1.5 times the total model input and 1.6 times the wall time in v19, and about 1.3 times both in v21.

The equal-information scavenger hunts also retain useful null results [58]. Both arms completed all sixteen runs of eight checkpoint-mode hunts, covering eighty stages each; only F001 required recovery after interruption, while the others finished within their first leg. In a separate equal-information relay hunt, fresh agents completed from their predecessors' written handoffs in both repetitions of both arms. Together these evaluations help distinguish accurate retrieval, economical reading, and sustained task execution rather than combining them into a single claim of improvement.

Equal-information evaluation	**Retrieval arm**	**Search arm**	**Measure and scope**
Recall bank, 300 questions	214/300	168/300	Correct memory identified; development/tuning bank, not a hunt.
Multi-day continuity v19	8/12	1/12	Both targets correct within 12,000 reading characters; partial Brain.
Padded stale-record v21	6/12	0/12	Both targets correct within 8,000 reading characters; partial Brain.
Eight checkpoint hunts, two repetitions	16/16; 80/80 stages	16/16; 80/80 stages	Same notes; interrupted recovery exercised only on F001.
Relay hunt, two repetitions	2/2	2/2	Fresh second agents completed from written handoffs.

5. Results, and the theory I want

What I care about in the end is the mathematics. AI systems did much of the mathematics in the following results, I directed the work and I am responsible for it, and each paper has a declaration of who did what.

They have been audited by independent agent runs and are not yet refereed. The agents worked from this memory, and I do not claim the memory caused any proof.

Ross [8] conjectured, in Leskelä's formulation [9], that a queue fed by a randomly varying arrival rate has no more mean workload when the environment moves faster. Take a rate-one single-server queue with Poisson arrivals of intensity $\varepsilon F(ct)$ and let $w_\varepsilon(c)$ be its stationary mean workload. The conjecture says w_ε is nonincreasing in c for every stationary ergodic F, and it is false. An explicit 64-state Markov intensity, with every transition rate positive and service times uniform on $[0.99, 1.01]$, has $w_\varepsilon(3\pi) - w_\varepsilon(2\pi) > \varepsilon^2/600$ for $0 < \varepsilon \le 10^{-4}$. The gap is proved only in light traffic and it is very small there. The second-order workload coefficient is nonincreasing in every bounded environment exactly when $s \mapsto \mathbb{E}(1 - \cos sB)/s^2$ is nonincreasing, for service time B [10].

In planar percolation, let f be the crossing function of a square of side R, $\mathcal{P}$ its pivotal set and $\mathcal{S}$ its spectral sample, the random set with law $\hat{f}(A)^2$. At the uniform measure the two sets have the same one- and two-point inclusion probabilities. For critical bond percolation on the square lattice and site percolation on the triangular lattice, the Shannon entropies $H(\mathcal{S})$ and $H(\mathcal{P})$ are both comparable to the total influence, which is of order $R^2\alpha_4(R)$ with α_4 the four-arm probability. The spectral bound answers the square-crossing question of Garban, Pete and Schramm [11, Section 11, Question 1], and the pivotal bound gives the other square-crossing bound in Kalai's Conjecture 8 [12]. It is a percolation case of the Fourier entropy-influence conjecture [13] and says nothing about general Boolean functions. Equal inclusion statistics and comparable entropy still do not make the sets alike. For critical triangular-site crossings the two laws are asymptotically mutually singular, which answers the discrete case of their Question 5 [11]: an elementary triangular face never lies in a pivotal set and appears with probability tending to one in the spectral sample, and the overlap on nonempty sets decays as $R^{-11/12+o(1)}$ [14]. This does not settle the continuum limits or the square-lattice bond case.

Vu conjectured [15, Conjecture 9.5] that an $n \times n$ random sign matrix M_n has all its eigenvalues in $\mathbb{Z}[i]$ with probability $2^{-(1+o(1))n^2}$. An explicit family gives $\Pr(\mathrm{Spec}(M_n) \subset \mathbb{Z}[i]) \ge 3^{\lfloor n/2 \rfloor} 2^{-\lfloor n^2/2 \rfloor}$, about the

square root of that, so the conjecture is false [16]. Lyons and Peres [17, Question 16.7] asked whether simple random walks on the base-2 and base-3 coding trees of the Cantor set give mutually singular harmonic measures on it, and they do [18]. For Aldous and Fill's Open Problem 7.30 [19], on every finite connected 1-walk-regular graph the heat-kernel ratio $H_t(x,y)/H_t(x,x)$, with $H_t = e^{-t(I-A/d)}$ on a d-regular graph, is strictly increasing in t across every edge [20]. In an excited random walk on the integers, the i-th visit to a site uses the i-th cookie of that site's stack, which biases the step. For IID stacks with nonnegative drift under weak ellipticity (Kosygina and Zerner [21, Problem 5.3]) the speed is zero when the expected total drift at a site is at most two and positive above two, with no bounded-stack assumption [22]. About 40 more write-ups of the same kind are on my GitHub. The brain itself is written up in [23].

The perceptron came earlier, in July, when the research brain was running and the math store did not exist yet, and two AI systems did most of the work under my supervision. Krauth and Mézard [24] predicted in 1989 that the largest number M_N of patterns the Ising perceptron stores at zero margin satisfies $M_N/N \to \alpha_\star \approx 0.833$. Ding and Sun [25] proved the lower bound and Huang [26] the upper bound, each conditional on a numerical condition about a low-dimensional variational problem that had been supported only by floating-point computation. My paper [27] verifies both outstanding conditions in certified interval arithmetic and re-derives the parameter rectangle, with $\alpha_\star \in [0.833078599,\ 0.833078600]$. This closes the conditional reduction, for Gaussian disorder and, by universality [28], for subgaussian entries such as ± 1. Huang's condition is a statement over the whole plane with a degenerate point. Reparametrizing by the moments of $(X, \tanh X)$ compresses the plane onto a compact convex body, and near that point, where no box enclosure can work, the proof certifies concavity of a majorant along rays. The certificate layer has a Lean 4 check in the repository [29]. The proof has passed a second-system audit and has not yet been independently reproduced.

My goal is to continue advancing mathematics, particularly probability theory and deep learning, while using those advances to improve the research process itself. Results about networks, algorithms, and computation can inform faster retrieval, more efficient memory, and better coordination within the Brain. I aim to create a feedback loop in which solving problems produces methods that strengthen the system, and those improvements help agents tackle harder mathematical and computational questions. This requires testing both the value of each result and whether incorporating it actually improves subsequent research.

I also want agents to communicate the ideas and solutions they develop, making research accessible through conversation and adaptive educational content. This motivates my work on representing mathematical ideas and generating explanations that respond to a person's questions. I envision someone calling a research agent to work through a paper, examine a proof in detail, or explore an unfamiliar concept, with the agent adjusting its explanation as the conversation develops. It could generate diagrams, slides, or a short video when a visual explanation would help, grounding these materials in the underlying sources and making uncertainty explicit. The goal is interactive, agentic content development that lets people question, understand, and participate in the research rather than simply receive its conclusions.

I already use this kind of interaction by calling a dot agent to read through papers, examine proofs and proposed solutions to open problems, and discuss how particular arguments developed. I want others to be able to do the same with an agent connected to the Brain, asking detailed questions and receiving explanations that draw on the research record and adapt as the conversation develops. This statement provides a concrete example: I asked a dot agent to retrieve my previous research statement, then worked through it over the phone, section by section and sentence by sentence. I explained my goals in my own words, asked the agent to propose wording and read it aloud, corrected its interpretations, and revised passages until they expressed what I intended. I also asked it to check numerical claims, find references, and incorporate recent results. This is the collaboration I want to develop further: people directing the discussion, questioning the material, and using the agent to deepen their understanding and express their ideas.

6. The next year

A year from now agents should run for much longer before they need me, on larger and harder problems, connecting areas of research that were not known to be connected. Recall has to get ten times faster and more exact. Section 3 took the easy part, not starting over on each call, and the rest has to come from mathematics. A floor on activation saved little on the separate walk that adds associations to a recall result and changed answers, so pruning that walk needs a bound on what it can change at the top of the list. Personalized PageRank has such a bound (the push algorithm of Andersen, Chung and Lang [30] keeps each entry within ε times its degree at a cost of order $1/(\varepsilon\alpha)$ for a seed of mass one, with α the restart probability, whatever N is), so a z-scored seed must first be clipped or split into positive and negative parts. The weight 0.3 came from a sweep on a development bank and the decay d was set by hand. Both can be fitted by maximum likelihood on a bank where we know the answer cards. The target is a tenth of the median latency on the same bank, with accuracy held within a point.

The bigger job is agency. An agent still chooses its next route by reading, and I want that choice to be a decision rule whose regret I can bound. Treat it as a bandit, where each route r has an unknown probability p_r of yielding a useful step and Thompson sampling [31] over Beta posteriors gives the order in which agents try routes. The record then becomes a posterior shared across agents and days. A route is dropped when the posterior probability that p_r exceeds a floor falls below a fixed level. The test the frontier layer needs is the same open problems with the record and without it, with useful steps per budget as the statistic (a blind test of results on 13 problems [1] was too small to tell). All my tests so far use one group's record, so I also want to repeat them on another group's, with its own agents and no shared history.

Which properties of these graphs predict that a brain will work? On the largest connected part of each graph, at a fixed time, P is a Markov chain, and I want a mixing quantity, the spectral gap or something local around the seed, that predicts the depth at which recall stops improving. Different types of centralities could choose what to serve. The topological and community structure of the mistakes may show where agents fail together. And do learned edges find connections nobody knew? Split the record by date and ask whether an edge between two fields learned before the split predicts a typed link or a result written after it. If it does, agents can follow those edges to new problems.

7. Random matrices and the stability of deep networks

The agents work on my other mathematics from this same memory. That mathematics started with my thesis under Leonid Berlyand, on which part of a trained network is signal and which is noise. Write a trained layer, an $N \times M$ weight matrix, as $W = S + R$, with S of low rank and R noise with independent mean-zero entries of variance σ^2/N and finite fourth moments. The Marchenko-Pastur law [32] says that as N grows with $c = M/N \le 1$ fixed, the singular values of R settle on a fixed density on $[\sigma(1-\sqrt{c}), \sigma(1+\sqrt{c})]$, so for a square matrix with $\sigma = 1$ a singular value at 3 is not noise. Berlyand, Sandier, Zhang and I [33] proved that when a layer has this form, with the singular values of S above the edge of this density (the bulk), for absolute-value activations, removing the singular values below that edge keeps the classification. With high probability an object classified with confidence above an explicit threshold is still classified correctly after the cut, which bounds the pruned accuracy from below. In our experiments on fully connected networks the cut increases accuracy. With Berlyand, Bourdais and Owhadi I carried this method to vision transformers and modern CNNs, using a regularizer that pushes the bulk and the outlying singular values (spikes) apart during training [34, 35]. My preprint on rectangular spikes treats the mesoscopic case, where the signal rank grows with N, and gives the outlier values and singular subspaces [36].

I also study whether a random network is stable at initialization. Take a square ReLU network of width n and depth L with independent $N(0, \alpha/n)$ weights, and let K be its global Euclidean Lipschitz constant, the largest operator norm of a Jacobian on a nonempty activation region. The answer is a transition at $\alpha = 2$, the value He initialization [37] uses. For each $\alpha < 2$ a sufficiently large fixed depth gives $K \le 1$ with exponentially high probability as $n \to \infty$, and for $\alpha > 2$ every fixed depth is expansive. For depth L let α_L be the variance below which that depth gives $K \le 1$. Then $\alpha_L \to 2$, with the certified lower bound $2\exp[-(\sqrt{10} + o(1))\sqrt{\log L/L}]$ [38]. The difficulty is that the region and its Jacobian depend on the same

weights. I handle this with coordinate sign changes that preserve the Gaussian law, together with Cover's hyperplane count [39], which cuts the entropy of the activation patterns from $nL\log 2$ to order $n\log L$. A recurrent network reuses one Gaussian W of entry variance $1/N$ at every step, $x_{t+1}=\mathrm{ReLU}(Wx_t)$, and the layers are no longer independent. There I prove exponentially concentrated norm propagation, $\|x_t\|^2\to 2^{-t}$, over every fixed time horizon for unit starts independent of W [40]. In the ordered phase, where the squared size of a small perturbation shrinks by a factor $\chi_1<1$ per layer, I have the exact constant for the smallest eigenvalue of the neural tangent kernel [41] of a deep tanh network at infinite width, for n distinct inputs on the fixed-point sphere, which sets the speed of gradient descent in the kernel regime: $\lambda_{\min}\sim\kappa\lambda_\perp L\chi_1^L$ [42]. I prove that the input-dependent factor $\lambda_\perp$ is positive, so the order Xiao, Pennington and Schoenholz [43] found for the non-constant eigenvalues is attained by the smallest one. With Berlyand, Sarapin, Slavin and Sodin [44] I studied the fixed points and the edge of chaos of randomly initialized autoencoders.

A few others. Under a uniform doubling condition on the data measure, accuracy cannot collapse while the cross-entropy keeps falling [45]. For random data with noisy labels, any two-layer network with m piecewise-linear neurons and arbitrary weights that fits below the noise level has Lipschitz constant at least of order $\sqrt{n/m}$ on n points, up to one logarithmic factor, which is the law of robustness conjectured by Bubeck, Li and Nagaraj [46]; see [47]. The smallest least singular value over all m-row submatrices of a tall Gaussian matrix has an exact exponential scale, and at the critical $N=2m-1$ real measurements this gives the sharp base $1/4$ in the Balan-Wang stability parameter [48] of phase retrieval [49]. With Werner Krauth (ENS Paris, manuscript not yet released [50]) I work on a diffusion sampler with no learned score, whose backward path is a mixture of Gaussian bridges to p sample points. Driven by the same noise, the paths $X^{(p)}$ and $X^{(\infty)}$ (from the true target) differ, on times away from the endpoint and for sub-Gaussian targets, by order $1/\sqrt{p}$, with an explicit Gaussian limit.

8. Kernels, and flows on the Bitcoin graph

At NASA's Jet Propulsion Laboratory in 2024 I worked on neural emulators of radiative transfer for imaging spectroscopy and on low-rank and sparse kernel methods. My later paper [51] uses a network g_θ for the mean plus a kernel regression on its residuals: $f(x)=g_\theta(x)+k(x,X)(k(X,X)+\lambda I)^{-1}(y-g_\theta(X))$. On public data, a Matérn kernel on the residuals and on learned features matches the best published result on structural mechanics [52] and improves on the published Gaussian-process emulator for OCO-2 [53] on its own test points, outright on two of three spectral bands. Next I want the agents working on better kernel methods, better kernels with DNNs, and improvements in regression, classification, pruning and adversarial robustness. The theorems of section 7 go in as linked cards, and the idea is to improve all of it while the brain is improving.

Bitcoin tracing is useful for following dirty money and memories are useful for building more agentic AI. Flows through Bitcoin and flows through memories have connections, which is why I study both. The question on the chain is what share of the value at a sink came from a given source, across tens of hops. On 96 real blocks my one-pass tracer handled all 161,524 sources with about 5,500 times less merge work than one trace per source, and 12 sources traced again separately matched to the satoshi. We have not traced a real criminal case. Value is conserved at each transaction, and with a nonnegative seed and damping for cycles personalized PageRank conserves mass too, so a card's score becomes the share of the question's mass that stops there, the same kind of quantity as a source's share at a sink.

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